

Sarah J. Bolton, Ph.D.

Major, United States Air Force
Assistant Professor, Department of Computer and Cyber Sciences
Director, Academy Center for Cyberspace Research
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PROFESSIONAL SUMMARY

Assistant Professor in the Department of Computer and Cyber Sciences at the United States Air Force Academy and Director of the Academy Center for Cyberspace Research. Teaches and mentors over 120 cadets per year across the computer science and cyber curriculum, with course director and vice chair responsibilities across the core sequence. Research and applied work span machine learning for aerospace and cyber data, computer science education, and human and machine teaming. Serves as dissertation chair and committee member for doctoral students conducting research at the intersection of AI, satellite communications, and space cybersecurity. Twenty four years of continuous Air Force service spanning intelligence, airborne mission operations, cyberspace operations, communications, and higher education.

EDUCATION

Doctor of Philosophy, Computer Science	2023
Air Force Institute of Technology, Wright Patterson AFB, Ohio	
<i>Research emphasis in artificial intelligence and machine learning</i>	
Master of Science, Computer Science	2018
Air Force Institute of Technology, Wright Patterson AFB, Ohio	
<i>Research focus on artificial intelligence and cognitive measurement</i>	
Bachelor of Science, Computer Science	2012
University of Arizona, Tucson, Arizona	
Associate of Applied Science, Communications Applications Technology	2008
Community College of the Air Force	
Associate of Arts, Spanish	2008
Defense Language Institute Foreign Language Center, Presidio of Monterey, California	

PROFESSIONAL MILITARY EDUCATION

Cyberspace 300, Wright Patterson AFB, Ohio	2026
Air Command and Staff College, Air University, Maxwell AFB, Alabama	2025
Cyberspace 200, Wright Patterson AFB, Ohio	2020
Squadron Officer School, Air University, Maxwell AFB, Alabama	2018
Undergraduate Cyber Training, Keesler AFB, Mississippi	2013
Airman Leadership School, Maxwell AFB, Alabama	2004

ACADEMIC APPOINTMENT

Assistant Professor and Director, ACCR

Sep 2023 to Present

Department of Computer and Cyber Sciences, United States Air Force Academy

- Director, Academy Center for Cyberspace Research (ACCR). Coordinate cyberspace research across the Academy, manage research collaborations and CRADAs, and connect cadets and faculty with external partners in government, industry, and academia.
- Coordinator, Cadet Summer Research Program (CSRP). Manage approximately 70 cadet summer research internships annually across DoD laboratories, federally funded research and development centers, and industry partners.
- ABET Lead for CS and Cyber Science Student Outcome 2 (design and evaluation of computing solutions).
- Course Vice Chair for CS220 (Data Structures and Systems Programming), CS380 (Design and Analysis of Algorithms), CS472 (Autonomous Systems Integration), CS314 (Quantum Computation), and CS426 (Languages and Machines).
- Led procurement and standup of the departmental AI computing cluster supporting cadet and faculty research.
- Faculty lead, CS and CyS Colloquium (CS405 and CyS405), curating speakers from across the Department of Defense, industry, government, and academia.

TEACHING

Courses Taught and Directed

- CS110, Introduction to Programming (Python). Foundations of programming, problem solving, and algorithmic thinking.
- CS210, Programming Fundamentals (C). Course Director. Programming fundamentals taught in C.
- CS220, Data Structures and Systems Programming. Course Director and Vice Chair. Arrays, lists, stacks, queues, trees, hash tables, and related systems concepts.
- CS380, Design and Analysis of Algorithms. Incoming Course Director and Vice Chair. Classic algorithm design paradigms and complexity analysis.
- CS426, Languages and Machines. Vice Chair. Compilers and automata theory.
- CS314, Quantum Computation. Vice Chair.
- CS472, Autonomous Systems Integration. Vice Chair.
- CS405 and CyS405, Computer Science and Cyber Science Colloquium. Senior level colloquium series.
- CS499 and Senior Colloquium support across the CS and CyS senior capstone sequence.

Teaching Development and Recognition

- Dean's Teaching Certificate, United States Air Force Academy.
- Course Design Institute, 2025.
- Mastery of Teaching Certification Community (MTCC), Cohort 8.

Teaching Philosophy

Emphasizes conceptual and framework based learning over syntax memorization, with frequent low stakes practice and feedback to build durable skill. Uses real world and operational examples to reinforce relevance and transfer, and provides clear structure, expectations, and scaffolding to support learners across a wide range of preparation. Course design is evidence based and systems oriented, with attention to genuine learning outcomes in a generative AI environment.

RESEARCH

Research interests span machine learning applied to aerospace and cyber data, cyber operations, computer science education, and human and machine teaming. Particular interest in how data and AI techniques can support operational decision making, and how learning experiences can be designed to produce adaptive technical leaders.

Research Areas

- Applied machine learning for aerospace data, including ADS-B classification and multi sensor aircraft identification.
- Cyber operations, network defense, and AI and ML in operational contexts.
- Computer science education, with a focus on retrieval practice, autograder design, and learning in a generative AI environment.
- Human and machine teaming and decision support.

DOCTORAL ADVISING

Dissertation Chair

Patrick Moore, Ph.D. Candidate

In Progress

Capitol Technology University

"Using AI to Mine Data and Provide Easily Interpretable Goal Oriented Insights: A Comparative Study"

Balasubramani Venugopal, Ph.D. Candidate

In Progress

Capitol Technology University

"Beyond the Horizon: The Future of Scalable and Secure Satellite Communications"

Dissertation Committee Member

Refat Zamkhshri, Ph.D.

2026

Capitol Technology University

"Satellite Ground Station Cybersecurity Challenges and Controls Effectiveness"

PUBLICATIONS

Peer Reviewed Journal Articles

Bolton, S., Dill, R., Grimaila, M. R., et al. (2023). ADS-B classification using multivariate long short term memory and fully convolutional networks and data reduction techniques. *The Journal of Supercomputing*, 79, 2281 to 2307.

Peer Reviewed Conference Proceedings

Bolton, S., Dill, R., Grimaila, M. R., and Hodson, D. D. (2023). Multi sensor aircraft classification. *2023 Congress in Computer Science, Computer Engineering, and Applied Computing (CSCE)*, Las Vegas, Nevada, pp. 796 to 800.

Bolton, S., Dill, R., Grimaila, M. R., and Hodson, D. D. (2023). Aircraft classification using flight phase identification. *NAECON 2023, IEEE National Aerospace and Electronics Conference*, Dayton, Ohio, pp. 192 to 197.

Under Review

Owen, L. R., Smith, J. B., Shin, E., and Bolton, S. (2026). Predictive modeling of G induced loss of consciousness (GLOC) using physiological data and machine learning. Submitted to *NAECON 2026, IEEE National Aerospace and Electronics Conference*.

Scholarship of Teaching and Learning

Bolton, S., Phan, T., and Crawl, M. (2026). Retrieval practice in action: Reflections on repeated testing in an introductory programming course. *USFAA Scholarship of Teaching and Learning Conference*.

MILITARY SERVICE

Twenty four years of continuous service in the United States Air Force, with operational, leadership, and academic assignments across the United States and deployments and missions in Colombia, Ecuador, Afghanistan, Qatar, and Djibouti. Entered active duty in August 2001 and commissioned through Officer Training School in March 2013.

Commissioned Assignments

Assistant Professor and Director, ACCR Department of Computer and Cyber Sciences, USAFA, Colorado	Sep 2023 to Present
Doctoral Student Air Force Institute of Technology, Wright Patterson AFB, Ohio	Aug 2020 to Sep 2023
Flight Commander, Network Operations 25th Space Range Squadron, Schriever AFB, Colorado <i>Deployed Apr 2019 to Nov 2019 as J6 Plans Officer, Combined Joint Task Force Horn of Africa, Camp Lemonnier, Djibouti</i>	Apr 2018 to Aug 2020
Master's Student Air Force Institute of Technology, Wright Patterson AFB, Ohio	Aug 2016 to Mar 2018
Officer in Charge, Network Control Center and Deputy Flight Commander Information Operations and Plans Flight, 90th Communications Squadron, F. E. Warren AFB, Wyoming <i>Deployed Jul 2014 to Jan 2015 as Deputy Systems Chief, Intelligence Systems Division, HQ USAFCENT, Al Udeid Air Base, Qatar</i>	Nov 2013 to Aug 2016
Undergraduate Cyber Training Student 333rd Training Squadron, Keesler AFB, Mississippi	Mar 2013 to Nov 2013
Officer Trainee, Officer Training School Maxwell AFB, Alabama	Jan 2013 to Mar 2013

Enlisted Assignments

Airman Education and Commissioning Program Student Air Force ROTC Detachment 020, University of Arizona, Tucson, Arizona	Aug 2010 to Jan 2013
Spanish Linguist, EC-130H COMPASS CALL Mission Crew Supervisor and Command Language Program Manager 55th Electronic Combat Group, Davis Monthan AFB, Arizona <i>Deployed Sep 2008 to Jan 2009, Afghanistan; mission support to Colombia and Puerto Rico</i>	Dec 2005 to Aug 2010
Spanish Linguist and RC-135 RIVET JOINT Analysis Operator 97th Intelligence Squadron, Offutt AFB, Nebraska <i>Augmented the Utah Air National Guard on Senior Scout missions in SOUTHCOM; deployments to Ecuador and Colombia</i>	Apr 2003 to Dec 2005
Combat Survival Training Fairchild AFB, Washington. SERE plus advanced and water survival training.	Jan 2003 to Mar 2003
Linguist Intelligence Training 316th Training Squadron, Goodfellow AFB, Texas	Jul 2002 to Jan 2003
Spanish Language Training Defense Language Institute Foreign Language Center, Presidio of Monterey, California	Sep 2001 to Jul 2002
Basic Military Training 323rd Training Squadron, Lackland AFB, Texas	Aug 2001 to Sep 2001

Flight Information

- Career Enlisted Aviator. 1,262.5 flight hours on EC-130H COMPASS CALL and RC-135 RIVET JOINT platforms.

Major Awards and Decorations

- Air Medal with one oak leaf cluster
- Aerial Achievement Medal with one oak leaf cluster
- Joint Service Commendation Medal
- Air and Space Commendation Medal with three oak leaf clusters
- Air and Space Achievement Medal with three oak leaf clusters
- Meritorious Unit Award with four oak leaf clusters
- Air and Space Outstanding Unit Award
- Combat Readiness Medal

PROFESSIONAL PROFILES

Google Scholar: scholar.google.com/citations?user=2_rtff4AAAAJ

ResearchGate: researchgate.net/scientific-contributions/Sarah-Bolton-2228137423

LinkedIn: linkedin.com/in/sarah-bolton-547a2019

Teaching site: sjbolton.github.io/teaching-site